

DATABASE DESIGN DOCUMENT

PART-4: BACKUP, RECOVERY, MIGRATION & DATABASE OPERATIONS

School ERP

PostgreSQL | Spring Data JPA | Hibernate

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Preamble

This document is Part-4, the final part, of the Database Design Document (DDD) v1.0 for the School ERP System, continuing directly from Parts 1-3. It defines PostgreSQL-specific operational standards for backup, recovery, disaster recovery, migration governance, environment configuration, monitoring, capacity planning, maintenance, and change management — the database-administration counterpart to the system-wide standards already fixed in MDB Part-4.

This document does not regenerate RGD v2.0, any prior Appendix, any UI/UX Screen Specification, or any prior MDB/DDD Part. It contains no SQL, no CREATE TABLE statements, no entity class definitions, no ER/UML diagram, no code, no scripts, and no cloud-vendor-specific content — every standard is stated as a rule only, applicable regardless of the eventual hosting provider.

Where this document's scope overlaps MDB Part-4 (Backup & Disaster Recovery Section 38, Production Operations Section 39), this DDD Part narrows that system-wide standard to its PostgreSQL-specific mechanics — it does not restate MDB Part-4's broader infrastructure scope, and it does not contradict any target already fixed there.

31. Backup Strategy

Fixes the PostgreSQL-specific mechanism realizing the Backup Strategy already defined in Appendix-L Section 12, Appendix-N Section 16, and MDB Part-4 Section 38.

- Continuous Write-Ahead Log (WAL) archiving runs at all times in every environment holding non-synthetic data (Staging, Production), supporting point-in-time recovery rather than restore-to-last-full-backup-only.
- A full base backup is taken daily, consistent with the Daily Full + Continuous Log standard already fixed in Appendix-F NFR-BKP-001 — the base backup and WAL archive together constitute the complete backup set.
- Backup artifacts are stored in a failure domain separate from the primary PostgreSQL instance, consistent with Appendix-L Section 12's requirement that a single infrastructure failure cannot destroy both the live data and its backup.
- Backup encryption at rest applies uniformly, consistent with MDB Part-2 Section 19 — a backup artifact is not exempt from the Encryption at Rest standard merely because it is not the live database.
- Redis, per DDD Part-1 Section 3 and MDB Part-3 Section 26, is backed up with lower rigor than PostgreSQL — cache data is reconstructable from PostgreSQL and is not treated as a system of record.

32. Restore & Recovery Standards

Fixes the verification and execution standard for restoring from the backups defined in Section 31, realizing the monthly restore-test requirement already fixed in Appendix-F NFR-BKP-005.

- A full restore-to-point-in-time is executed against a non-Production environment on a monthly schedule — a backup that has never been restored is not considered verified, consistent with Appendix-L Section 12.1.
- Restore procedures are documented and version-controlled alongside the migration standards (Section 34 this document), so a restore is a rehearsed, repeatable operation, not an improvised response to an incident.
- Recovery Time Objective and Recovery Point Objective for a restore operation match the targets already fixed in Appendix-F NFR-BKP-003/004 — this document does not set new targets, only the PostgreSQL-specific procedure for meeting them.
- A restore operation is logged with actor, timestamp, and outcome, extending the audit discipline already fixed in DDD Part-3 Section 24 to infrastructure-level database operations, not only application-level ones.

33. Disaster Recovery Standards

Fixes the PostgreSQL-specific failover mechanism realizing the Disaster Recovery flow already defined in Appendix-L Section 13 and Appendix-N Section 15.

- The streaming-replication read replica (DDD Part-1 Section 3) is the primary failover target — replica promotion to primary is the first recovery action for a primary-instance failure, ahead of restoring from backup (Section 32), consistent with the recovery priority already fixed in Appendix-N Section 15.1.
- Replica promotion is rehearsed at the same annual DR-drill cadence already fixed in Appendix-F NFR-DR-002, executed in the Staging environment (Appendix-N Section 5) as part of that environment's designated DR-rehearsal role.
- Following promotion, a new replica is provisioned to restore the primary/replica topology before the DR event is considered fully resolved — operating on a promoted-but-unreplicated primary is treated as a continuing risk state, not a resolved one.
- Application Layer reconnection to the newly-promoted database instance follows the same flow already fixed in Appendix-L Section 13 — this DDD Part does not redefine that application-side behavior, only confirms the database-side promotion it depends on.

34. Database Migration Governance

Extends the Database Migration Standards already fixed in MDB Part-3 Section 28 with the formal governance/approval workflow.

- Every migration is authored against the logical model already approved in Appendix-H — a migration introducing an element not traceable to Appendix-H or a formally logged Schema Extension item (per the UI/UX Screen Specifications' flagging convention) requires an explicit, documented exception before it can be approved.
- Migration review requires two approvals: the standard Code Review Checklist (MDB Part-3 Section 30, DDD Part-3 Section 30) and, per DDD Part-2 Section 18, sign-off from the Business Owner of any table the migration affects.
- Migrations progress through the same environment sequence already fixed in Appendix-L Section 4.1 (Development → Testing → UAT → Staging → Production) — a migration is never applied to Production without having first been applied, and verified, in Staging against production-representative data volume.
- The specific migration tool remains Client/Product Decision Required, as already noted in MDB Part-3 Section 28 — this governance workflow applies regardless of which tool is ultimately selected.

35. Environment Database Standards

Fixes database-specific configuration per environment, extending Appendix-N Section 5/9.1 and Appendix-M Section 12 (Test Data Management).

- Development and Testing run a single, non-replicated PostgreSQL instance with synthetic data only, consistent with Appendix-N Section 9.1 — no environment other than Staging and Production runs the full primary/replica topology (Section 33).
- UAT and Staging use anonymized, production-representative data, per Appendix-M Section 12's synthetic/anonymized test-data principle — Sensitive/PII fields (DDD Part-2, Appendix-G/H Sensitive Data Matrix) are never populated with real values outside Production.
- Training (Appendix-N Section 5) uses synthetic, reset-friendly data specifically designed to be safely repeatable across many rehearsal sessions, distinct from the sign-off-bearing UAT dataset.
- Database credentials (MDB Part-2 Section 33) differ per environment without exception — no credential used in Production is ever reused in a lower environment.

36. Monitoring & Health Standards

Fixes PostgreSQL-specific metrics extending the Monitoring Matrix already established in Appendix-L Section 21.1 and the general Monitoring & Health Checks already fixed in MDB Part-4 Section 37.

Metric Category	PostgreSQL-Specific Signal
Availability	Primary instance reachability; replica connectivity and streaming status
Replication Health	Replication lag (time and byte-based), tied directly to the RPO target (Section 32)
Query Performance	Slow-query log activity against the $\leq 200\text{ms}$ indexed-query target (Appendix-F NFR-PERF-012)
Connection Pool	HikariCP pool utilization (DDD Part-1 Section 3) per Application Layer instance
Storage Growth	Table and partition size trend, feeding Capacity Planning (Section 37 this document)
Lock Contention	Deadlock and long-held-lock detection, particularly around the pessimistic-locking cases already identified in MDB Part-3 Section 25 (e.g., seat allocation)
Maintenance Health	Autovacuum/statistics currency (Section 38 this document)

37. Capacity Planning Standards

Fixes the ongoing capacity-review discipline against the Data Volume Estimates already fixed in Appendix-H Section 17.7 and reused throughout this documentation set.

- Storage and compute capacity are reviewed at each academic-session boundary (DDD Part-2 Section 12 principle), not on a purely reactive, alert-triggered basis, since enrollment and transaction volume step-change at year start rather than growing smoothly.
- Partition growth (Section 26, DDD Part-3) is monitored against the estimates already fixed in Appendix-H Section 17.7 (10 million+ AttendanceRecord/year, 15 million+ AuditLog/year estimated, 500,000+ Invoice/Payment/year) — a partition materially exceeding its estimate triggers a capacity review, not merely a storage-alert.
- The 5-year growth horizon already fixed in Appendix-H Section 17.6 and Appendix-I Section 18.2 remains the planning target — capacity decisions are evaluated against this horizon, not only against the immediate next academic session.
- Read-replica capacity (Section 33) is reviewed independently of primary-instance capacity, since Reports-module query load (Appendix-K Section 18) grows differently than transactional write load.

38. Database Maintenance Standards

Fixes routine PostgreSQL housekeeping required to sustain the Performance Standards already fixed in DDD Part-3 Section 27.

- Autovacuum remains enabled and actively tuned for the high-volume, high-write tables already identified in DDD Part-2 Section 11 (Transaction Data classification) — autovacuum is never disabled system-wide as a performance workaround.
- Table and index statistics are kept current for the query planner, with particular attention to the partitioned high-volume tables (DDD Part-3 Section 26), since stale statistics on a large partitioned table are a common, otherwise-silent cause of query-plan regression.
- Index health (bloat, unused-index detection) is reviewed on a scheduled basis, informing whether an index already recommended in Appendix-H Section 12 remains justified as data volume and query patterns evolve.
- Maintenance windows for any operation with a performance or availability impact follow the same planned-maintenance-window standard already fixed in Appendix-L Section 14 and Appendix-N Section 13 — routine maintenance is not performed unannounced during academic operating hours.

39. Database Change Management

Fixes the approval and execution discipline for any Production database change, extending Separation of Duties already fixed in Appendix-L Section 22.2 and MDB Part-4 Section 39.

- No individual holds both migration-approval authority (Section 34) and unrestricted Production database execution access — this mirrors the same Separation of Duties principle already applied to Release approval versus Production access in Appendix-L Section 22.2.
- Every Production database change — schema migration, configuration change, manual data correction — is logged with actor, timestamp, and business justification, whether or not it originates from an automated migration pipeline (Section 34) or a declared incident response (MDB Part-4 Section 39).
- An emergency change (executed outside the standard migration governance workflow, Section 34, due to an active incident) requires retroactive review and sign-off within a defined window — an emergency change is never left permanently unreviewed.
- Database Administrator access to Production is itself reviewed periodically against the Role structure already fixed in Appendix-I Section 8 and Appendix-L Section 22, consistent with the access-revocation discipline already fixed for staff exit (BR-HR-002).

40. Database Operations Checklist

The final consolidation point of the Database Design Document, verified on an ongoing operational cadence and before every major release, extending the Database Performance Review Checklist already fixed in DDD Part-3 Section 30 and the Production Readiness Checklist in MDB Part-4 Section 40.

#	Checklist Item
1	Backup (Section 31) completed successfully within the last 24 hours; WAL archiving current
2	Monthly restore test (Section 32) passed within the last calendar month
3	Replica (Section 33) healthy, streaming, with replication lag within target
4	No pending migration (Section 34) awaiting Business Owner or Code Review sign-off
5	Environment data (Section 35) confirmed free of real Sensitive/PII values outside Production
6	Monitoring signals (Section 36) confirmed flowing and alert-routed correctly
7	Capacity review (Section 37) current for the active academic session
8	Autovacuum and statistics health (Section 38) within normal range for all high-volume tables
9	No unreviewed emergency change (Section 39) outstanding
10	Database Performance Review Checklist (DDD Part-3 Section 30) passed for any pending release

A negative finding against any item above is treated as a release-blocking condition for Production promotion, consistent with the Quality Gate discipline already fixed in Appendix-M Section 25 and the Production Readiness Checklist in MDB Part-4 Section 40 — this checklist is the database-specific instantiation of that same governing principle, not a separate or lesser standard.